Answer key: Lesson 6, The School Shop

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Ben has exactly $10. He buys two different things. Look at the four tags.

Answer: $3 and $7

3 needs 7 to make 10, and there is a $7 tag. 5 needs 5, but there is only one $5 tag and the two things must be different. 8 needs 2, and there is no $2 tag. So the two tags are $3 and $7.

Watch for: Using the $5 tag twice.; Choosing $3 and $5 or $5 and $7 without adding.

S2. Siti buys a ruler for $7 and a cup for $8. How much does she pay altogether?

Answer: $15

7 and 8: 7 and 3 make 10, and 5 more makes 15. Siti pays $15 altogether.

Watch for: Taking away: 8 take away 7 is 1.; Writing $78 by joining the digits.

S3. Wei has $20. He buys a jar for $14. How much money does he have left?

Answer: $6

Count up from 14 to 20. That is 6 steps. Wei has $6 left.

Watch for: Adding 20 and 14.; Writing 14, the price of the jar.

S4. Ali has $10. A ruler costs $7 and a glue stick costs $5.

Answer: No; $2

7 and 5 make 12. 12 is more than 10, so $10 is not enough. Count up from 10 to 12: Ali needs $2 more.

Watch for: Saying Yes because each price is less than 10.; Writing 12 for the money needed.

S5. A ruler costs $7. Mei buys 3 rulers.

Answer: $7 + $7 + $7 = $21; $21

7 and 7 make 14. 14 and 7 make 21. Mei pays $21 for 3 rulers.

Watch for: Writing 7 or 14 (one or two rulers).; Writing $73 by joining digits.

S6. Fill in the blanks in the table.

Answer: 8; 9; 17

Ravi: 20 take away 12 is 8. Aisha: she had 15 and has 6 left, so she spent 9. Kumar: he spent 8 and has 9 left, so he had 17.

Watch for: Adding when the question needs take away.; Writing the same number in two rows.

Think harder (page 3)

T1. Exactly $20. Pens cost $2 each and books cost $7 each. Ali spends exactly $20 on pens and books. He may buy 0 books.

Answer: 10; X; 3; X; 2 ways

0 books: $20 is left, which is 10 pens. 1 book: $13 is left, and $2 pens cannot make 13 (X). 2 books cost $14. 20 take away 14 is 6, which is 3 pens. 3 books cost $21, which is too much (X). So there are 2 ways: 10 pens, or 2 books and 3 pens.

Watch for: Stopping after the first way.; Counting 2 books and 3 pens and 3 pens and 2 books as two ways.

T2. A child in another class lists ways to pay exactly $16 with $2 notes and $5 notes. The list says: eight $2 notes.

Answer: No; 2 five-dollar notes and 3 two-dollar notes

One $5 note leaves $11, which $2 notes cannot make. Two $5 notes leave $6, which is 3 two-dollar notes. So the missing way is 2 five-dollar notes and 3 two-dollar notes. Three $5 notes are $15 and leave $1, which cannot be made.

Watch for: Saying Yes because 8 two-dollar notes work.; Writing 2 five-dollar notes and 6 two-dollar notes.

T3. Ravi buys a book for $9, a hat for $4 and a pen for $2. He pays with a fifty-dollar note.

Answer: $15; $35

9 and 4 make 13. 13 and 2 make 15. Count up from 15 to 20: 5. Then from 20 to 50: 30. 5 and 30 make 35. Ravi gets $35 change.

Watch for: Adding only two of the prices.; Writing 15 as the change.

T4. Rubbers cost $3 each and pencils cost $4 each. Ali spends exactly $18. Tick True or False.

Answer: True; True; False; False

6 rubbers cost 18, so it is a way. 2 rubbers and 3 pencils cost 6 and 12, which is 18, so it is a way. The same things in a different order are the same way. 4 rubbers and 2 pencils cost 12 and 8, which is 20, not 18.

Watch for: Saying a different order is a new way.; Not adding the cost of both things.

Go further (page 4)

G1. Pay exactly $20 with $10 notes, $5 notes and $2 notes. Write how many of each note. Find every way. Way 1 is shown. Any order of the ways is fine.

Answer: Six ways: (2, 0, 0), (1, 2, 0), (1, 0, 5), (0, 4, 0), (0, 2, 5), (0, 0, 10) as ten-dollar, five-dollar, two-dollar notes

Two $10 notes. One $10 note leaves $10: two $5 notes, or five $2 notes. No $10 note: four $5 notes, or two $5 notes and five $2 notes, or ten $2 notes. That is 1 and 2 and 3, which is 6 ways.

Watch for: Stopping at the ways with a $10 note.; Counting the same notes in a different order twice.

G2. Ali buys 7 things. Stickers cost $2 each and pads cost $5 each. He spends exactly $20.

Answer: 5 stickers; 2 pads

Try 1 pad: 6 stickers cost 12, and 12 and 5 make 17. Too small. Try 2 pads: 5 stickers cost 10, and 10 and 10 make 20. So 5 stickers and 2 pads.

Watch for: Buying 7 of the same thing.; Giving a way with more or fewer than 7 things.

G3. Write an exactly puzzle for a partner. Choose two of these tags. Many answers are right.

Answer: Many answers are right: the two prices must add to the amount

For example, the pencil $3 and the cup $8: spend exactly $11 on two things. The two things cost $3 and $8. Check that the two prices add to the amount.

Watch for: Writing an amount that is not the total of two tags.; Choosing the same tag twice.

Investigation (page 5): Amounts you cannot pay

First: yes, one $5 note and one $2 note. Mi cannot pay $1 or $3. Every other amount from $2 to $12 can be paid: $2, $4, $5, $6, $7, $8, $9, $10, $11 and $12.

Look for: A child who tries each amount from $1 up with a list of notes; a child who sees that $1 and $3 are too small to be made; a child who notices that from $4 on every amount works. Credit any clear method. Do not mark a child down for stopping early.

Exit ticket (page 5)

E1. Mei spends exactly $11 on two things. The tags say $3, $4, $5 and $8. Which two did she buy? Choose the correct answer. Write its number in the bracket.

Answer: (2) $3 and $8

3 and 8 make 11. 3 and 4 make 7, 4 and 5 make 9, and 5 and 8 make 13.

Watch for: Choosing (4) $5 and $8 because they are the biggest.; Choosing a pair without adding.

E2. Ben has $12. He wants a pen for $6 and a book for $8. Does he have enough money?

Answer: No

6 and 8 make 14. 14 is more than 12, so $12 is not enough.

Watch for: Saying Yes because each price is less than 12.; Taking away: 8 take away 6 is 2.

E3. A cup costs $3. Ali buys 4 cups. How much do the cups cost altogether?

Answer: $12

3 and 3 make 6. 6 and 3 make 9. 9 and 3 make 12.

Watch for: Writing 3 or 7.; Counting only 3 cups.

Sneak-peek puzzles (not in the workbook)

1. $5 (3 and 2 make 5).
2. $2 and $7, or $4 and $5. Both pairs make $9: 2 and 7 make 9, and 4 and 5 make 9.